Agent Generated Hypothesis and Research Plan

Novel Hypothesis:

Metformin, in combination with a glutaminase inhibitor (e.g., CB-839), synergistically disrupts metabolic plasticity in Glioblastoma Multiforme (GBM) cells, preventing their adaptation to nutrient stress and leading to enhanced cell death compared to either agent alone. This synergistic effect stems from metformin's AMPK activation and reduction of glucose metabolism, rendering GBM cells more vulnerable to glutaminase inhibition and disruption of glutamine-dependent pathways.

Research Plan:

- **Cell Culture Models:**
- Utilize a panel of established human GBM cell lines (e.g., U87-MG, A172, LN229) representing different genetic subtypes (e.g., EGFR-amplified, PTEN-deleted, IDH-wildtype).
- Establish patient-derived GBM cell lines to increase translational relevance and capture inter-patient heterogeneity. Culture these lines in serum-free media supplemented with growth factors (EGF, FGF) to maintain stem-like properties.
- **Metabolic Assays:**
- **Glucose and Glutamine Uptake Assays:** Measure glucose and glutamine uptake rates in GBM cell lines treated with metformin, CB-839, or the combination. Use commercially available kits to quantify glucose and glutamine concentrations in the culture media.
- **Lactate Production Assay:** Quantify lactate production as a marker of glycolysis in treated and untreated cells.
- **ATP Measurement:** Assess intracellular ATP levels as a measure of overall cellular energy status. Decreased ATP levels upon combined treatment would suggest a metabolic catastrophe.
- **Oxygen Consumption Rate (OCR) and Extracellular Acidification Rate (ECAR):** Perform Seahorse XF analysis to assess mitochondrial respiration (OCR) and glycolysis (ECAR) in real-time. This will provide a comprehensive view of metabolic fluxes.
- **Metabolomics:** Perform untargeted metabolomics analysis on cell lysates from each treatment group to identify alterations in metabolic pathways beyond glucose and glutamine metabolism. This could reveal novel metabolic vulnerabilities induced by the combination treatment.
- **Cell Viability and Apoptosis Assays:**
- **MTT or MTS Assay:** Measure cell viability after treatment with metformin, CB-839, and the combination at various concentrations and time points.
- **Annexin V/Propidium Iodide (PI) Staining:** Quantify apoptosis and necrosis using flow cytometry.
- **Caspase-3/7 Activity Assay:** Measure caspase-3/7 activity as an indicator of apoptosis.
- **Colony Formation Assay:** Assess the long-term effects of the treatment on cell survival and proliferation by measuring colony-forming ability.
- **Western Blot Analysis:**
- Assess the expression and phosphorylation status of key proteins involved in metabolic regulation, such as AMPK, ACC, mTOR, and glutaminase.

- Examine downstream targets of AMPK to confirm its activation by metformin.
- Monitor the expression of apoptosis-related proteins (e.g., cleaved caspase-3, PARP) to confirm apoptosis induction.
- **In Vivo Studies:**
- **Orthotopic Xenograft Model:** Implant GBM cell lines (e.g., U87-MG) intracranially into immunocompromised mice (e.g., NSG mice).
- Treat the mice with vehicle control, metformin, CB-839, or the combination.
- Monitor tumor growth using MRI or bioluminescence imaging.
- Assess survival rates in each treatment group.
- Perform immunohistochemistry on tumor tissue to assess proliferation (Ki-67), apoptosis (cleaved caspase-3), and metabolic markers (e.g., glutaminase expression).
- **Patient-Derived Xenografts (PDX):** Implant tumor fragments from GBM patients into immunocompromised mice. This will provide a more clinically relevant model to test the efficacy of the combination therapy.
- **Statistical Analysis:**
- Use appropriate statistical tests (e.g., t-tests, ANOVA) to compare the effects of different treatments on cell viability, metabolic parameters, and tumor growth.
- Perform synergy analysis using CalcuSyn software to determine whether the combination of metformin and CB-839 exhibits synergistic effects.

Novelty Assessment:

Based on the PubMed searches conducted, the core hypothesis of synergistically targeting metabolic plasticity in GBM with a combination of metformin and a glutaminase inhibitor appears reasonably novel. While several studies have investigated metformin's effects in GBM and others have explored glutaminase inhibition, the specific combination and the emphasis on disrupting metabolic plasticity as the primary mechanism of synergy do not seem to be extensively studied based on the PubMed search results. The search results revealed studies on metformin's anti-tumor effects, its ability to enhance immunotherapy, and its potential to induce senescence in GBM cells. They also highlighted the importance of metabolic reprogramming in GBM. However, none of the retrieved articles explicitly investigated the synergistic effect of metformin and a glutaminase inhibitor on metabolic plasticity disruption in GBM. Therefore, this hypothesis warrants further investigation.